Introductory activity

Activity 1 · An "impossible" computation that gives the right answer 40 min . In the sixteenth century, a formula was known for solving equations of the form $x^3 = px + q$. It is given here and taken for granted:

$$x = \sqrt[3]{\frac{q}{2} + \sqrt{\frac{q^2}{4} - \frac{p^3}{27}}} + \sqrt[3]{\frac{q}{2} - \sqrt{\frac{q^2}{4} - \frac{p^3}{27}}}.$$

We study the equation $x^3 = 15x + 4$.

1. Check that 4 is a solution of the equation.
2. Compute $\frac{q^2}{4} - \frac{p^3}{27}$ for $p = 15$ and $q = 4$. What becomes of the formula? Can it be applied in $\mathbb{R}$?
3. A calculator of the time decides to carry on as if a number written $\sqrt{-1}$ existed, with the single rule $\sqrt{-1} \times \sqrt{-1} = -1$ and the usual rules of computation for everything else. Expand $\left(2 + \sqrt{-1}\right)^3$ with these rules.
4. Assume that $\sqrt{-121}$ can be written $11\sqrt{-1}$. Show that the first term of the formula can be taken equal to $2 + \sqrt{-1}$, then find in the same way a value for the second term.
5. What does the formula then give? Compare with question 1.
6. Is this computation a proof? Which rules would have to be fixed for it to become one?